

Purple Team Outline

Week 03: Network Protocol Abuse

Objective: To simulate and detect availability-based attacks and traffic interception (Man-in-the-Middle).

Task 01: Denial of Service (DoS) Simulations

- **Description:** Understand how to disrupt service availability and how to monitor for such disruptions.
- **Technical Requirements:**
 - o Differentiate between DoS and DDoS architectures in a technical report.
 - o Simulate a DoS attack against a vulnerable network service (e.g., SYN Flood or Resource Exhaustion).
- **Learning Goal:** Identify the point of failure in network protocols when stressed beyond their limits.
- **What to Document :**
 - o Install **hping3** in your Kali etc.
 - o Perform TCP SYN Flood (**Dos**) attack on the vulnerable target machine with your actual **IP** – Attach the overwhelmed service screenshot.
 - o Perform Icmp Flood (**Dos**) attack on your vulnerable target machine with the random **IP** – Attach the overwhelmed service screenshot.
 - o Open **tcpdump/wireshark** & analyze all the packets – Attach the captured packets screenshot.
 - o In your own words, explain how a system can be protected or defended against DoS attacks such as SYN Flood and ICMP Flood.
 - o Do you think DoS/DDoS attacks will remain a major threat for large organizations in 2026 and the coming years? If yes, recommend possible solutions or strategies that companies can implement to mitigate these threats.

Task 02: MITM & ARP Spoofing Analysis

- **Description:** Intercept traffic between two legitimate nodes on the local network.
- **Technical Requirements:**
 - Execute an ARP Spoofing attack to position the attacker machine as the "Man-in-the-Middle."
 - Use Wireshark to capture and analyze redirected packets, identifying unencrypted credentials or data.
- **Learning Goal:** Understand the inherent trust vulnerabilities in ARP and the importance of encryption (TLS/SSH).
- **What to Document :**
 - Install Ettercap on Kali Linux and identify the version you are using – Attach a **screenshot showing the installation and the Ettercap version**.
 - Using the **sysctl** tool, configure your Kali machine to forward packets across the network by setting: **net.ipv4.ip_forward = 1** & explain why this step is required for packet forwarding and **MITM** attacks – Attach a screenshot showing the command execution and output.
 - Determine and display the **IP addresses of both vulnerable machines and the attacker machine** – Attach a **screenshot showing the IP addresses of all three systems**.
 - Start Ettercap in **CLI mode** to intercept communication between the two victim machines, at the same time, run **Wireshark or tcpdump** in the background to capture and analyze network traffic. Attach the following screenshots:
 - Ettercap running in CLI mode
 - Wireshark capturing packets during the attack
 - From **Victim Machine 1**, send any message to **Victim Machine 2** using **curl** on **port 80 (HTTP)**. Analyze the captured packets in Wireshark to observe the transmitted data. Use this step to understand the **difference between HTTP (port 80) and HTTPS (port 443)** – Attach a **packet capture screenshot showing the transmitted message**.

- Connect both victim machines to a **VPN** and repeat the interception process using Ettercap & Capture the traffic using **Wireshark** and analyze the packets.Explain:
 - What differences do you observe between the earlier packets and the packets captured after connecting to the VPN & Why those differences occur ?

Attach screenshots of: Victim machines connected to the VPN & the captured packets in Wireshark

- In your own words, explain what you learned from **Task 2 (MITM attack demonstration)**. Also describe **how systems and users can defend against MITM attacks**.
- Do you believe **Man-in-the-Middle (MITM) attacks** are a serious threat for people using **public Wi-Fi networks** (such as in cafes or restaurants)? Explain your reasoning and provide recommendations for users to stay secure on public networks.